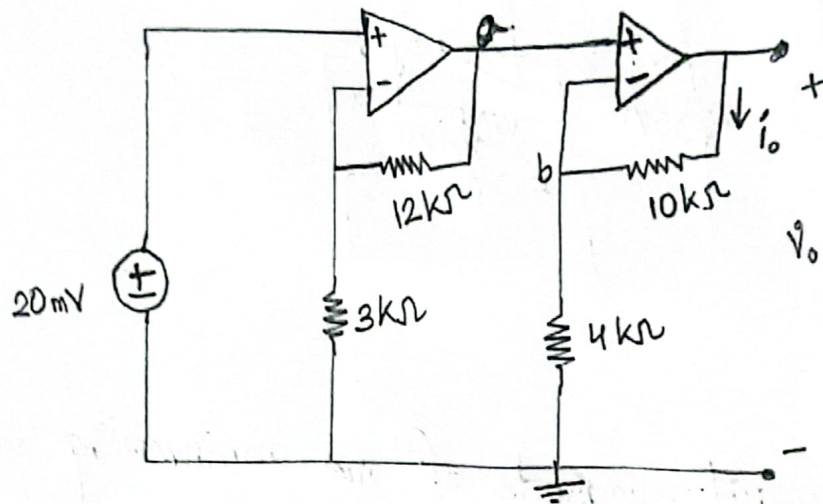


Topic:- Op-amp

1] calculate V_o and i_o



Solution -

The circuit consists of two - noninverting amplifiers.

At the output of first op-amp,

$$\begin{aligned} V_a &= \left(1 + \frac{R_f}{R_i}\right) \times V_{in} \\ &= \left(1 + \frac{12}{3}\right) \times 20 \\ &= 100 \text{ mV} \end{aligned}$$

$$\begin{aligned} R_f &= 12 \text{ k}\Omega \\ R_i &= 3 \text{ k}\Omega \\ V_{in} &= 20 \text{ mV} \end{aligned}$$

At the output of second op-amp,

$$\begin{aligned} V_o &= \left(1 + \frac{10}{4}\right) V_a \\ &= 350 \text{ mV} \end{aligned}$$

$$\begin{aligned} R_f &= 10 \text{ k}\Omega \\ R_i &= 4 \text{ k}\Omega \\ V_a &= 100 \text{ mV} \end{aligned}$$

(Answer)

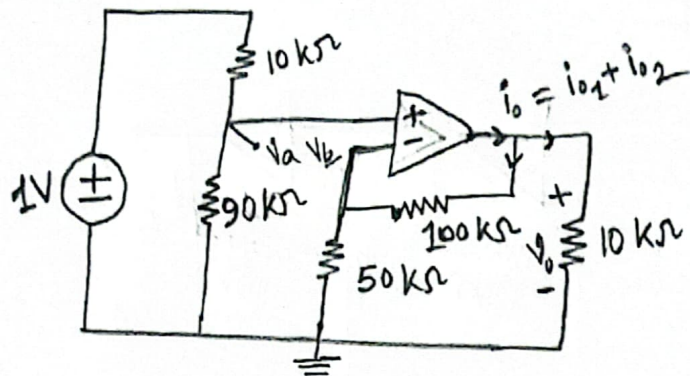
$$\begin{aligned} \text{Now, } i_o &= \frac{V_o - V_b}{10} \\ &= \frac{(350 - 100)}{10} \text{ mA} \\ &= 0.025 \text{ mA} \end{aligned}$$

But $V_b = V_a$

$$\therefore \left[\frac{1 \text{ mV}}{1 \text{ k}\Omega} = 1 \text{ }\mu\text{A} \right]$$

(Answer)

2] . calculate V_o and i_o :-



Solution :-

The circuit consists of one noninverting amplifier.

$$V_a = \left(\frac{90}{10+90} \right) \times V_{in}$$

$$= \frac{9}{10} \times 1$$

$$= 0.9V$$

$$\therefore V_a = V_b = 0.9V.$$

$$V_o = \left(1 + \frac{R_f}{R_i} \right) \times V_a$$

$$= \left(1 + \frac{100}{50} \right) \times 0.9$$

$$= 2.7V. \quad \text{(Answer)}$$

i_o is divided into two parts i_{o1} and i_{o2}

$$i_{o1} = \left(\frac{2.7 - 0.9}{100} \right)$$

$$= 0.018 \text{ mA}.$$

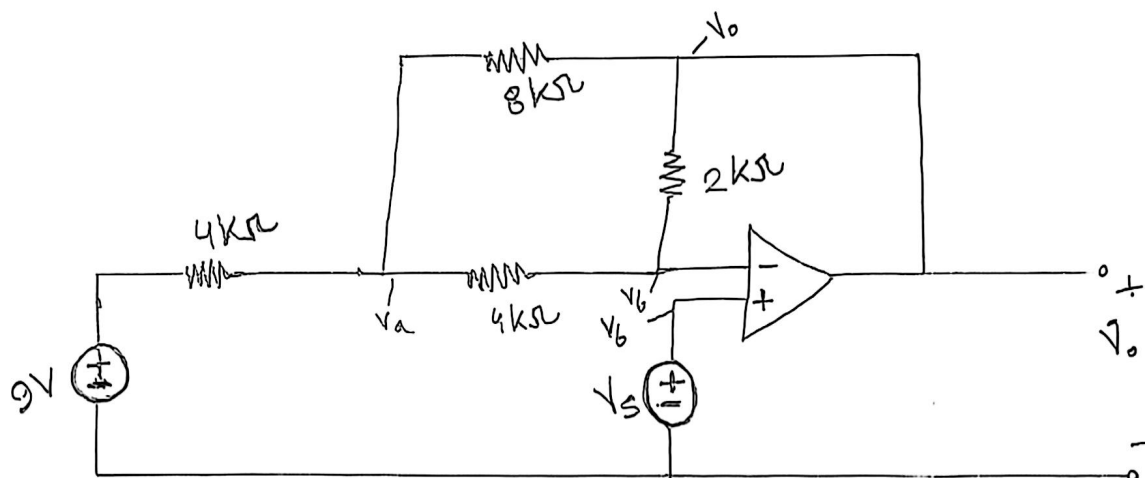
$$i_{o2} = \frac{2.7 - 0}{10}$$

$$= 0.27 \text{ mA}.$$

$$\text{Now, } i_o = (0.018 + 0.27) \text{ mA}$$

$$= 0.288 \text{ mA}. \quad \text{(Answer)}$$

3] calculate V_o while $V_s = 2V$ and $V_s = 0V$.



Solution :-

For $V_s = 2V$, $[V_b = V_s = 2V]$

At node a,

$$\frac{V_a - 9}{4} + \frac{V_a - V_o}{8} + \frac{V_a - V_b}{4} = 0.$$

$$\Rightarrow \frac{2V_a - 18 + V_a - V_o + 2V_a - 2V_b}{8} = 0.$$

$$\Rightarrow 5V_a - V_o = 18 + 2 \times 2$$

$$\Rightarrow 5V_a - V_o = 22 \quad \text{--- (I)}$$

At node b,

$$\frac{V_b - V_a}{4} + \frac{V_b - V_o}{2} = 0.$$

$$\Rightarrow \frac{V_b - V_a + 2V_b - 2V_o}{4} = 0.$$

$$\Rightarrow -V_a - 2V_o = -(2 + 2 \times 2)$$

$$\Rightarrow -V_a - 2V_o = -6 \quad \text{--- (II)}$$

solving the equations,
we get,

$$V_a = \frac{50}{11} V$$

$$V_o = \frac{8}{11} V$$

Answer

For $V_s = 0V$

$$V_s = V_b = 0V$$

At node a,

$$\frac{V_a - 9}{4} + \frac{V_a - V_o}{8} + \frac{V_a - 0}{4} = 0$$

$$\Rightarrow \frac{2V_a - 18 + V_a - V_o + 2V_a}{8} = 0$$

$$\Rightarrow 5V_a - V_o = 18 \quad \text{--- (I)}$$

At node b,

$$\frac{0 - V_a}{4} + \frac{0 - V_o}{2} = 0$$

$$\Rightarrow \frac{-V_a - 2V_o}{4} = 0$$

$$\Rightarrow -V_a - 2V_o = 0 \quad \text{--- (II)}$$

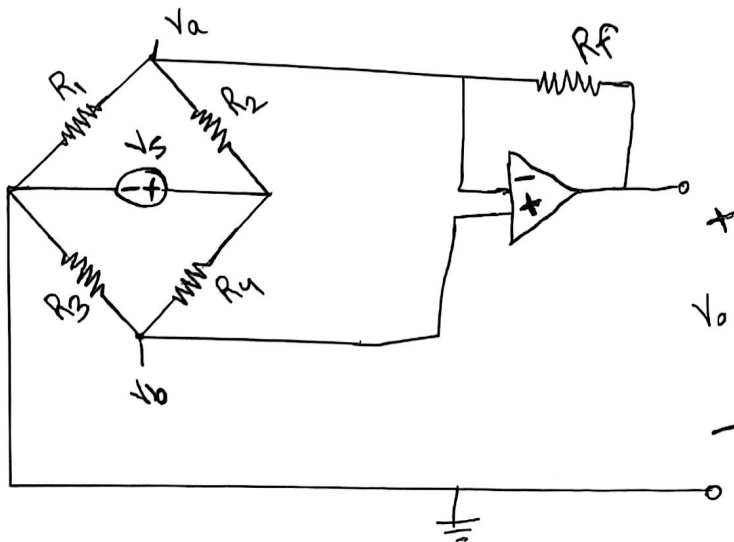
Solving the equations,

$$V_a = \frac{36}{11}V$$

$$V_o = -\frac{18}{11}V.$$

Answer

4) calculate V_o while $V_s = 10V$



Solution :-

using KVL,

$$V_a = \frac{R_1}{R_1 + R_2} \times V_s$$

$$= \frac{1}{2} \times 10$$

$$= 5V$$

$$V_b = \frac{R_3}{R_3 + R_4} \times V_s$$

$$= 5V$$

let, $R_1 = R_2 = R_3 = R_4 = 1k\Omega$

Ideal case

Now, $V_o = (5 - 5)V$
 $= 0V$

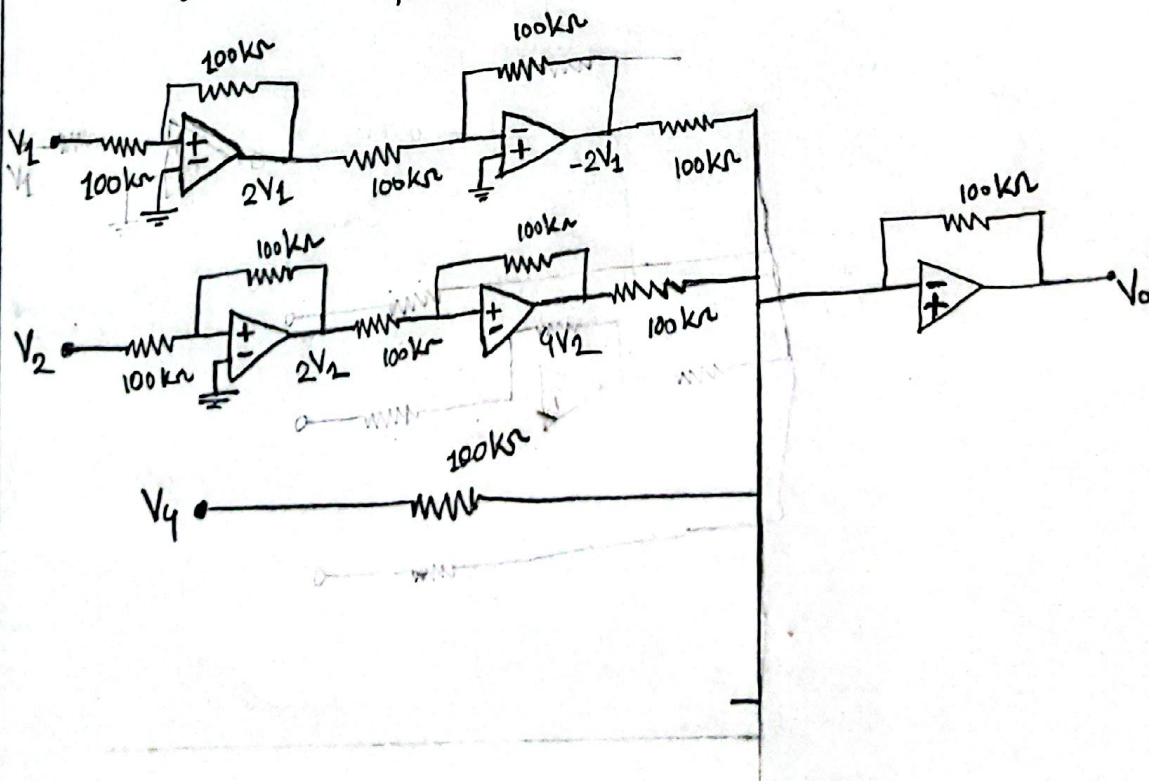
Answer}

5/ Design an op-amp for $V_o = 2V_1 - 4V_2 + 5V_3 - V_4$ All resistance must be $100k\Omega$

Solution :-

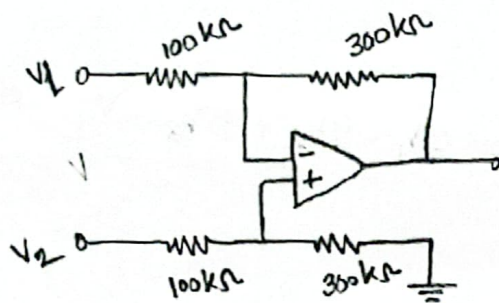
Its a summing Amplifier,

$$V_o = -\frac{R_f}{R_1} \cdot V_1 - \frac{R_f}{R_2} \cdot V_2 - \frac{R_f}{R_3} \cdot V_3 - \frac{R_f}{R_4} \cdot V_4$$

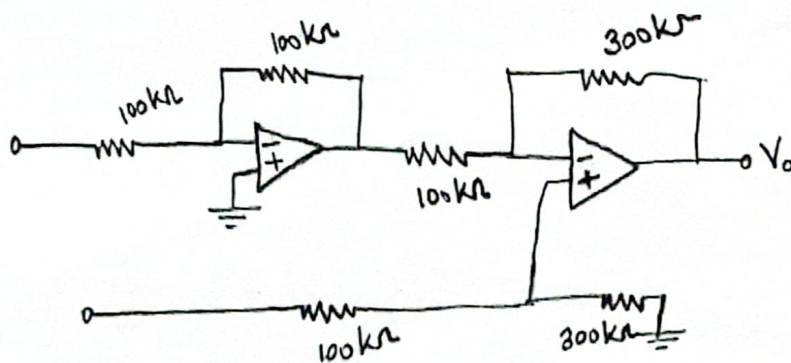


6]. Design a circuit to amplify the difference between two inputs by 3. (a) use only one opm (b) use two opms.

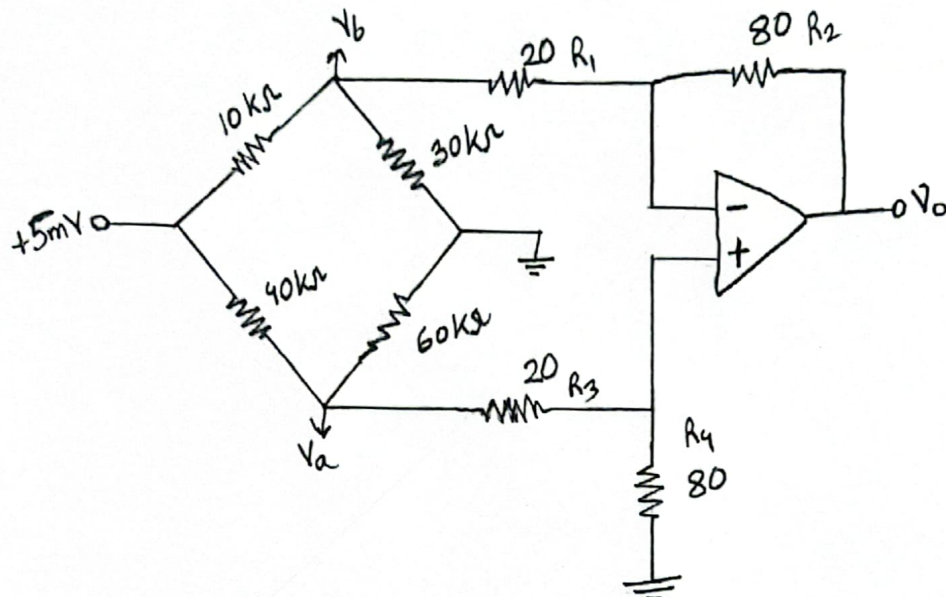
a



b



7.] Calculate V_o



Solution :-

Differential Amplifier V_o — Gain $(V_a - V_b)$

$$V_o = 0 \Rightarrow \text{condition, } \frac{R_2}{R_1} = \frac{R_4}{R_3}$$

$$V_o = \frac{R_2}{R_1} (V_2 - V_1) = \frac{R_4}{R_3} (V_2 - V_1)$$

$$V_b = \frac{30}{30+10} \times 5 \text{ mV}$$

$$= 3.75 \text{ mV}$$

$$V_a = \frac{60}{60+40} \times 5 \text{ mV}$$

$$= 3 \text{ mV}$$

Now,

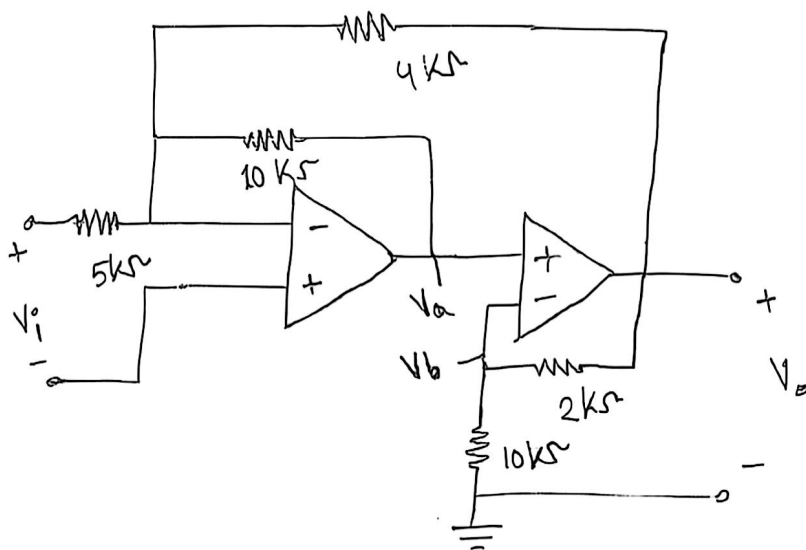
$$V_o = \frac{80}{20} (V_a - V_b)$$

$$= 4 (3 - 3.75)$$

$$= -3 \text{ mV.}$$

(Answer)

8]. Calculate V_o :-



Solution :-

$$V_a = -\frac{10}{5} \cdot V_i - \frac{10}{4} \cdot V_o$$

$$\Rightarrow V_a = -2V_i - 2.5V_o \quad \text{--- (1)}$$

combining the equations,

$$\frac{5}{6}V_o = -2V_i - 2.5V_o$$

$$\Rightarrow V_o \left(\frac{5}{6} + 2.5 \right) = -2V_{in}$$

$$\Rightarrow V_o = \frac{-2 \times 3}{10} V_{in}$$

$$\therefore V_o = -0.6V_{in}$$

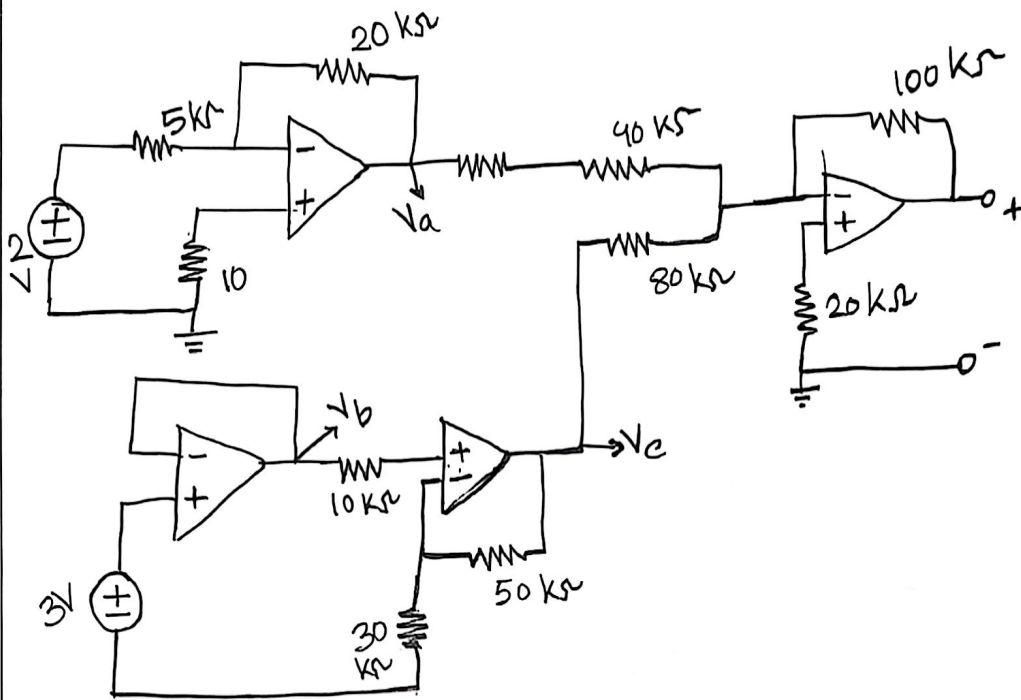
$$[V_a = V_b]$$

$$V_a/V_b = \frac{10}{10+2} \times V_o$$

$$= \frac{5}{6} V_o \quad \text{--- (1)}$$

(Answer)

9] calculate V_o



Solution :-

$$V_a = -\frac{20}{5} \times 2 = -8V.$$

$$V_b = 3V. \quad [\text{Buffer Amplifier } V_{in} = V_{out}]$$

$$V_c = \left(1 + \frac{50}{30}\right) \times 3$$

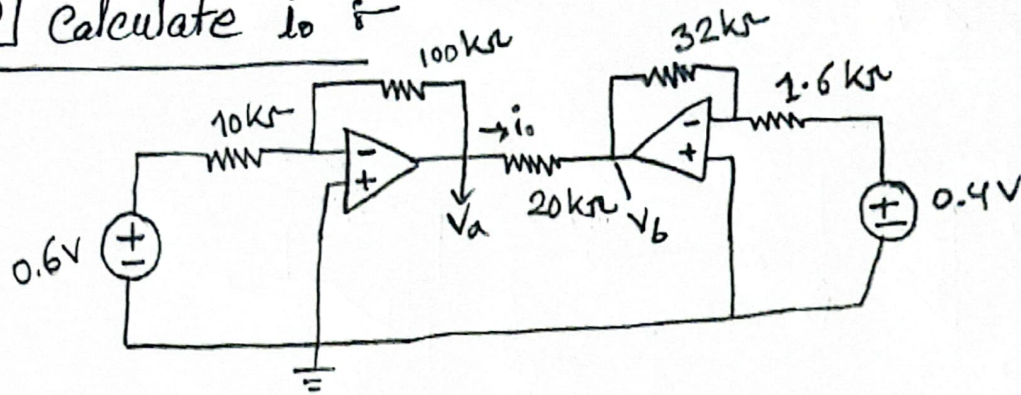
$$= 8V.$$

$$\text{Summing Amplifier } V_o = -\frac{100}{40} \times (-8) - \left(\frac{100}{80} \times 8\right)$$

$$= 20 - 10$$

$$= 10V$$

10/ calculate i_o :-



Solution :-

$$\text{For } V_a = \left(\frac{-100}{10} \times 0.6 \right) \\ = -6V$$

$$\text{For } V_b = \left(\frac{-32}{1.6} \times 0.4 \right) \\ = -8V$$

$$\text{we know, } i_o = \frac{V_a - V_b}{20k\Omega} = \frac{-6 - (-8)}{20k\Omega} \\ = 0.1 \text{ mA} \quad \text{(Answer)}$$